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Daniel Marín Pina

Research activities

I am a theoretical astrophysicist based in the University of Heidelberg, Germany. My main interest is in stellar and black hole dynamics in star clusters, and the formation of dormant star-black hole binaries and gravitational waves from binary black hole mergers. My goal for the next years is to develop models for the population of these dormant star-black hole binaries, in preparation for the next *Gaia* data release and results from other massive surveys. Furthermore, I want to keep working on modelling the dynamical formation of binary black hole mergers in star clusters. This will help us understand the production of eccentric gravitational waves and the seeding and growth of intermediate-mass black holes.

Education

PhD Cum Laude in Astronomy and Astrophysics — Barcelona, Spain
Institute of Cosmos Sciences, University of Barcelona — 2021 - 2025

- Thesis title: Black hole dynamics in star clusters: Eccentric mergers and star-black hole binaries
- Grade: Excellent Cum Laude (maximum possible grade)
- Advisor: Prof. Mark Gieles

Master in Astrophysics, Particle Physics and Cosmology — Barcelona, Spain
University of Barcelona — 2019 - 2020

- Master's thesis title: Event horizon in a compact binary coalescence
- Advisor: Prof. Roberto Emparan

Bachelor in Physics — Barcelona, Spain
University of Barcelona — 2015 - 2019
NTNU Norwegian University of Science and Technology — 2018

Work experience

DFG Postdoctoral Fellow — Heidelberg, Germany
Institute for Theoretical Astrophysics, University of Heidelberg — 2025 - 2028 (ongoing)

FPI Predoctoral Fellow — Barcelona, Spain
Institute of Cosmos Sciences, University of Barcelona — 2021 - 2025

***Gaia* DPAC Postgraduate Researcher** — Barcelona, Spain
Technological Unit, Institute of Cosmos Sciences, University of Barcelona — 2019 - 2021

Awards and Honors

- **PhD Thesis Cum Laude** — University of Barcelona
  "Black hole dynamics in star clusters: Eccentric mergers and star-black hole binaries" 2025
- **Best Poster Presentation** — ESO Garching, Germany
  "Two in a million: The interplay between binaries and star clusters" 2023
- **Best Master's Thesis** — University of Barcelona
  "Event horizon in a compact binary coalescence" 2020

Teaching and advising

1. Coadvising Master student Bruno Rando Forastier — 2023-2024
2. Coadvising Bachelor student Clàudia Garcia Diago — 2024-2025
3. Teaching assistant: Algebra and lineal geometry (12h) — 2024-2025
4. Teaching assistant: Algebra and lineal geometry (12h) — 2023-2024
5. Teaching assistant: Algebra and lineal geometry (30h) — 2022-2023
6. Teaching assistant: Calculus I (15h) — 2021-2022

Other scientific work

Journal referee

- Refereed for Astronomy and Astrophysics
- Refereed for Astrophysics and Space Science
- Refereed for Physical Review D

Local organising committee Institute for Theoretical Astrophysics, 2026

- Member of the LOC for the conference "AMUSE Spring School 2026"

Scientific organising committee Institute of Space Sciences, 2024

- Member of the SOC for the conference "Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles 2024"

PhD representative University of Barcelona, 2022 - 2025

- I contributed to set up a transversal predoctoral community in my local institution (∼70 PhD students), and acted as a 'PhD representative'
- Our goal was to establish a network for informal collaboration, enhance our transversal skills and knowledge and create a safe space to fight against any and all forms of harassment.
- My role as a PhD representative was to organise regular workshops on transversal skills such as public speaking, outreach tips and tricks, Git courses for first-years and career in astronomy.

Short author list publications

1. Dynamical formation of dormant black holes: Predictions for Gaia DR4
   Daniel Marín Pina, Michela Mapelli, Giuliano Iorio and Cecilia Sgalletta, arXiv:2610.18601 (2026), submitted to A&A
2. Fast Dynamical Modelling of Milky Way Globular Clusters – II. Impacts of Black Hole Prescriptions
   Nolan Dickson, Vincent Hénault-Brunet, Fotios Fronimos Pouliasis, Mark Gieles, **Daniel Marín Pina** and Peter J. Smith, arXiv:2609.10955 (2026), submitted to ApJ
3. N-body modelling of the ED-2 stream progenitor shows Gaia BH3's formation involved dynamical interactions
   Daniel Marín Pina, Mark Gieles, Sara Rastello, Clàudia Garcia-Diago, Giuliano Iorio and Marc Ardèvol, A&A, 712, A161 (2026)
4. clusterBHBdynamics: A fast code for the evolution of tidally limited star clusters and their binary-black-hole mergers
   Fotios Fronimos Pouliasis, Nolan Dickson, **Daniel Marín Pina**, Mark Gieles, Vincent Hénault-Brunet and Fabio Antonini, A&A, 713, A111 (2026)
5. The role of accretion efficiency, natal kicks, and angular momentum transport in the formation of the Gaia black holes
   Michela Mapelli, Cecilia Sgalletta, Johanna Müller-Horn, Giuliano Iorio, Stefano Rinaldi, Christian Burt, **Daniel Marín Pina** and Amedeo Romagnolo, A&A, 713, A58 (2026)
6. Repopulating the pair-instability mass gap without sustained growth to massive IMBHs: the case of 47 Tuc
   Debatri Chattopadhyay, **Daniel Marín Pina**, Mark Gieles, Fabio Antonini and Fotios Fronimos Pouliasis, arXiv:2604.09773 (2026), submitted to MNRAS
7. The formation of Gaia BH_3
   Daniel Marín Pina, Mark Gieles, Sara Rastello and Giuliano Iorio, arXiv:2510.07021 (2025), submitted to MODEST-25 IAU Conference proceedings
8. Interactions among binary black holes in star clusters: Eccentric gravitational wave captures and triple formation
   Daniel Marín Pina, Mark Gieles, Tomas Andrade and Alessandro A. Trani, A&A, 698, A229 (2025)
9. Dynamical formation of Gaia BH_3 in the progenitor globular cluster of the ED-2 stream
   Daniel Marín Pina, Sara Rastello, Mark Gieles, Kyle Kremer, Laura Fitzgerald and Bruno Rando Forastier, A&A, 688, L2 (2024)

10. Demographics of three-body binary black holes in star clusters: implications for gravitational waves
Daniel Marín Pina and Mark Gieles, MNRAS, 527, 8369–8381 (2024)

11. Binary-single interactions with different mass ratios: Implications for gravitational waves from globular clusters
Bruno Rando Forastier, **Daniel Marín Pina**, Mark Gieles, Simon Portegies Zwart and Fabio Antonini, A&A, 697, A118 (2025)

12. Event horizon of a charged black hole binary merger
Daniel Marín Pina, Marta Orselli and Daniele Pica, PhRvD, 106, 084012 (2022)

13. Precursory collapse in neutron star-black hole mergers
Roberto Emparan and **Daniel Marín Pina**, PhRvD, 102, 024009 (2020)

Large collaboration publications

- 17 refereed papers in the Gaia Collaboration
- 27 refereed papers in the Virgo Collaboration

Grants received

- **FPI predoctoral grant** ($\sim$ 100 000€) — Barcelona, Spain
  Spanish ministry of Education and Professional Formation — 2021-2025
- **Summer research contract** ($\sim$ 1000€) — Barcelona, Spain
  Institute of Space Studies of Catalonia (IEEC) — Summer 2019
- **Collaboration grant** (2000€) — Barcelona, Spain
  Spanish ministry of Education and Professional Formation — 2019
- **Erasmus+** ($\sim$ 1000€) — Trondheim, Norway
  European Commission — 2019

Skills

Languages: English (C2), Spanish (Native), Catalan (Native), Japanese (B1), Chinese (B1).

Programming: `Python`, `Mathematica`, `Bash`, `Java`, `Fortran`, `C/C++`.

Specific software: `PeTar`, `NBODY6++`, `Tsunami`, `clusterBHBdynamics`.

Talks and posters

1. Contributed talk: "Bridging binary stellar evolution and cluster dynamics: modelling the formation of Gaia BHs with PeTar + SEVN"
MODEST 26: Modelling and Observing Dense Star Clusters
ICISE, Quy Nhon, Vietnam — August 20, 2026

2. Invited seminar: "The dynamical origin of Gaia BH_3"
MPIA Galaxy Coffee
Max Planck Institute for Astronomy, Heidelberg, Germany — December 11, 2025

3. Contributed talk: "The formation of Gaia BH_3"
The Lifecycle of Stellar Black Holes
Kavli Institute for Theoretical Physics, Santa Barbara, USA — November 18, 2025

4. Seminar: "The dynamical origin of Gaia BH_3"
STRAND Theory Meeting
University of California San Diego, San Diego, USA (online due to transportation issues)
December 6, 2025

5. Contributed talk: "The dynamical origin of Gaia BH_3"
IAU Symposium 398/MODEST-25: Compact Stars and Binaries in Dense Star Clusters
Seoul National University, Seoul, South Korea — June 15–20, 2025

6. Contributed talk: "The dynamical origin of Gaia BH_3"
IAU Symposium 398/MODEST-25: Compact Stars and Binaries in Dense Star Clusters
Seoul National University, Seoul, South Korea — June 15–20, 2025

7. Contributed talk: "The dynamical origin of Gaia BH_3"
MODEST-25SPZ: Advanced Simulations and Modeling in Astrophysics
Academiehuis Zwolle, Zwolle, the Netherlands — May 19–23, 2025

8. Invited seminar: "Interactions among binary black holes in star clusters"
   Cardiff Astro Seminar series
   Cardiff University, Cardiff, United Kingdom — January 22, 2025

9. Contributed talk: "BBH-BBH interactions: eccentric gravitational wave captures and triple formation"
   New ideas on the origin of black hole mergers
   Niels Bohr International Academy, Copenhagen, Denmark — August 12–16, 2024

10. Contributed talk: "Where do gravitational waves with non-negligible eccentricity come from?"
    Cross-Collserola PhD Meeting in Astrophysics, Cosmology and Particles 2023
    Institute of Cosmos Sciences, Barcelona, Spain — October 6, 2023

11. Contributed talk: "Dynamical BBH formation and disruption: Implications for gravitational waves"
    MODEST-23: Star clusters in the post-pandemic era
    Northwestern University, CIERA, Evanston, USA — August 28-September 1, 2023

12. Contributed talk: "The demographics of dynamically-formed BBHs in star clusters"
    Nordic Winter School on Gravitational Astrophysics
    Niels Bohr International Academy, Skeikampen, Norway — January 29-February 3, 2023

13. Contributed talk: "Black hole mergers from star clusters"
    IEEC Forum
    Institute of Space Studies of Catalonia, Barcelona, Spain — June 2, 2022

14. Invited talk: "Black hole mergers from star clusters"
    ICCUB Winter Meeting
    Institute of Cosmos Sciences, Barcelona, Spain — February 7-8, 2022

15. Poster: "Intermediate-mass black holes in Milky Way globular clusters: theoretical expectations from growth modelling"
    MODEST 26: Modelling and Observing Dense Star Clusters
    ICISE, Quy Nhon, Vietnam — August 17-21, 2026

16. Poster: "Dynamical BBH formation and disruption: implications for gravitational waves"
    Two in a million: The interplay between binaries and star clusters
    European Southern Observatory (ESO), ESO Garching, Germany — September 11-15, 2023

17. Poster: "The demographics of dynamically-formed BBHs in star clusters"
    SEA XV Scientific Meeting
    Spanish Astronomical Society, La Laguna, Spain — September 5-9, 2022

Selected outreach

1. A dance of stars and black holes
   Outreach activity for girls
   Girl's day – Heidelberg Universität, Heidelberg, Germany — April 23, 2025

2. Espectrografia de forats negres i validacions de relativitat general amb GW250114
   LVK Outreach translation (to Catalan)
   LVK Collaboration, Online — September 9, 2025

3. Passeig virtual per la galàxia
   Outreach activity for kids
   Centre cívic Vil·la Urània, Barcelona, Spain — May 26, 2025

4. Passeig virtual per la galàxia
   Outreach activity for kids
   Centre cívic Vil·la Urània, Barcelona, Spain — May 14, 2025

5. Interview #MotorDeLaRecerca
   Instagram interview
   @ubdivulga, Barcelona, Spain — April 29, 2025

6. Universitat de l'Experiència – L'Univers com un tot: L'univers en contínua expansió I, II, III
   Outreach lecture series for senior citizens (6h)
   Universitat de Barcelona, Barcelona, Spain — April 1, 8 and 22, 2025

7. Telenotícies - Gran alineament dels planetes
   TV interview
   TV3 - National Catalan TV, Barcelona, Spain — February 28, 2025

8. Universitat de l'Experiència – L'Univers com un tot: L'univers en contínua expansió I, II, III
   Outreach lecture series for senior citizens (6h)
   Universitat de Barcelona, Barcelona, Spain — April 2, 16 and 18, 2024

9. Experiments de Física 2024
   Outreach lecture for high school students (8h)
   Universitat de Barcelona, Barcelona, Spain — January 23 and 31, 2024

10. L'Univers és ple de colors: l'astrofísica a través de múltiples mirades
    Outreach lecture series
    Biblioteca Gabriel García Márquez, Barcelona, Spain — September 26, 2023

11. Il·luminem els forats negres: iniciació a l'astrofísica i la gravitació
    Outreach lecture series (12h)
    Centre Cívic Casa Golferichs, Barcelona, Spain — April 27-June 29, 2023

12. Constel·lacions en 3D
    Outreach activity for kids
    Biblioteca Municipal de Bellavista, Les Franqueses del Vallès, Spain — February 4, 2023

13. Forats negres amb Dani Marín
    Radio interview
    La Caravana de Ràdio Estel, Barcelona, Spain — February 1, 2023

14. Experiments de Física 2023
    Outreach lecture for high school students (8h)
    Universitat de Barcelona, Barcelona, Spain — January 23 and 25, 2023

15. Apadrina un doctorand Roast Battle (Encontres tercer cicle)
    Outreach lecture for undergraduate students
    Universitat de Barcelona, Barcelona, Spain — November 15, 2022

16. Constel·lacions en 3D
    Outreach activity for kids
    15a festa de la ciència, Barcelona, Spain — May 29, 2022

17. Il·luminem els forats negres: iniciació a l'astrofísica i la gravitació
    Outreach lecture series (12h)
    Centre Cívic Casa Golferichs, Barcelona, Spain — April 28-June 16, 2022

18. Mirem la lluna des de la plaça de la palmera
    Astronomical observation
    Pla d'Acció Comunitària de la Verneda i la Pau – Associació ÀGORA, Institut de Ciències del Cosmos de la Universitat de Barcelona, Barcelona, Spain — June 10, 2022

19. Desenvolupament de la carrera professional
    Round table
    IEEC Forum 2022, Barcelona, Spain — June 2, 2022

Conferences

1. MODEST 26: Modelling and Observing Dense Star Clusters
   ICISE
   Quy Nhon, Vietnam — August 17-21, 2026

2. Stellar-Mass Black Holes at the Nexus of Optical, X-ray, and Gravitational Wave Surveys
   Kavli Institute for Theoretical Physics
   Santa Barbara, California, USA — November 3-December 4, 2025

3. The Lifecycle of Stellar Black Holes
   Kavli Institute for Theoretical Physics
   Santa Barbara, California, USA — November 17-20, 2025

4. IAU Symposium 398/MODEST-25: Compact Stars and Binaries in Dense Star Clusters
   Seoul National University
   Seoul, South Korea — June 15–20, 2025

5. MODEST-25SPZ: Advanced Simulations and Modeling in Astrophysics
   Academiehuis Zwolle
   Zwolle, the Netherlands — May 19–23, 2025

6. Challenges and future perspectives in gravitational- wave astronomy: O4 and beyond
   Lorentz Center
   Leiden, the Netherlands — October 14–18, 2024

7. New ideas on the origin of black hole mergers
   Niels Bohr Institute
   Copenhagen, Denmark — August 12–16, 2024

8. Two in a million: The interplay between binaries and star clusters
   European Southern Observatory (ESO)
   ESO Garching, Germany — September 11-15, 2023

9. MODEST-23: Star clusters in the post-pandemic era
   Northwestern University, CIERA
   Evanston, USA — August 28-September 1, 2023

10. XV Scientific Meeting
    Spanish Astronomical Society (SEA)
    Tenerife, Spain — September 5-9, 2022

Schools

1. AMUSE Spring School 2026
   Institute for Theoretical Astrophysics
   Heidelberg, Germany — March 23-March 27, 2026

2. Nordic Winter School on Gravitational Astrophysics
   Niels Bohr International Academy
   Skeikampen, Norway — January 29-February 3, 2023

3. Workshop on Gravitational Wave Modelling
   University of Barcelona
   Barcelona, Spain — October 10-11, 2022

4. Gravitational Wave Open Data Workshop
   University of Heidelberg
   Heidelberg, Germany (Online) — May 23-25, 2022

5. N-Body Simulation Lecture
   University of Heidelberg
   Heidelberg, Germany (Online) — September 27-29, 2021

6. Summer School on Gravitational Wave Astrophysics
   Niels Bohr International Academy (NBIA)
   Copenhagen, Denmark (Online due to COVID) — August 2-6, 2021